

$f \mapsto \frac{\partial}{\partial x} f$ and $f \mapsto \int_a^b f(x) dx$ are both linear

notation when we write $x \mapsto \text{some expression involving } x$

We say this is an anonymous function

$$x \mapsto x^2 \text{ and } f(x) = x^2$$

Q: if we know L is linear (but don't know the formula for L) how much info do we need to know about L to compute its output for every input

$$L: P_2 \mapsto P_0 \rightarrow \text{"R"} \quad L(x^2 + 3x) = ?$$

$$\begin{aligned} L(ax^2 + bx + c) &= L(ax^2 + bx) + L(c) \\ &= L(a[x^2 + \frac{b}{a}x]) + L(c) \\ &= aL(x^2 + \frac{b}{a}x) + L(c) \end{aligned}$$

in this ex. this isn't enough info

Q: if $L: A \rightarrow B$ what must be true about A and B ?

OK: differentiable funcs, integrable functions, $\mathbb{R}$, points in $\mathbb{R}^2$

not OK: points on the sphere (S^2)

Points in $\mathbb{R}^2$ can be scaled ex: $(x, y) \mapsto (13x, 13y)$

- ① if $x \in A$ then $ax \in A$
② there is a set of scalars that we can pull out of linear functions

ex: P_2 has scalars from $\mathbb{R}$

the set of scalars must be a field $\mathbb{C}$ are also a field

③ if $x, y \in A$ then $x+y \in A$

④ $+$ in A is

- commutative $x+y = y+x$

- associative $(x+y)+z = x+(y+z)$

- has a 0 (additive identity)
 $0+x = x$

- has negatives (additive inverse)
 $x+(-x) = 0$

⑤ distributive properties

$$a(x+y) = ax+ay \quad \text{or} \quad (a+b)x = ax+bx$$

a set satisfying properties 1, 2, 3, 4, 5 is a vector space

P_2 : check ①-⑤

$$4 - \sqrt{\cdot}, \sqrt{\cdot}, f(x) = 0, \sqrt{\cdot}$$

2 - scalars are $\mathbb{R}$

$$5 - \sqrt{\cdot}, \sqrt{\cdot}$$

1 - closed under scalar multiplication

3 - closed under $+$

